When Can a Swarm Be Audited?[1]

Evidence limits and an evaluation protocol

Razvan Sacara | Swarm Alignment Hackathon participant

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Abstract

Multi-agent oversight requires evidence connecting an action to the instance that executed it. A plausible account can remain unsupported when that connection is missing. This scoping study identifies a feasible evaluation of executor attribution under selective evidence loss and changing agent names. We review the sprint materials, selected past winners, primary papers and the supplied workshop transcripts. Existing work already covers trace tampering, evidence binding, external execution receipts, distributed attacks and failure attribution; none of these is claimed as our invention. We propose a narrower stress test that crosses a fixed export-record deletion policy with identity churn while preserving hidden collector originals. An elementary two-history construction shows why identical observations cannot identify different executors, even for a stronger reader. The prospective design compares readers on identical packets, separates identity bindings from execution receipts, and distinguishes unsupported accusations from supported definite findings and justified abstention. One primary contrast is specified at 25% loss, with a separate interaction analysis and episode-level uncertainty. No agent-performance experiments are reported. The contribution is a source-grounded task definition, a bounded information limit and a falsifiable protocol. Whether selective loss produces a practically important multi-agent effect remains to be measured.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

Suppose two workers can update the same fake record. An exported log says that the update completed, but omits the link between the call and its worker. An investigator may suspect one worker without being able to establish which instance acted. This example separates three questions: whether a policy violation occurred, who executed the mutation, and whether several agents colluded. Only the first two have operational labels in this proposal. A shared bad outcome does not establish intent or collusion [1,2,23].

We ask whether selective loss of coordination-relevant export records impairs supported executor attribution more than random loss, especially when display names change. The target is the executor of one designated unauthorized mutation, not the agent whose earlier mistake caused the failure. This narrower question fits oversight and agent-infrastructure research without requiring a general-purpose swarm runtime.

The completed work is a scoping synthesis and an elementary information-limit argument. The experiment is prospective. Our proposed contribution is the interaction between a declared deletion policy, changing names and separately retained identity/execution evidence. Novelty, practical value and effectiveness are not yet established.

2. Related Work

Qin et al. already study trace deletion, peer influence and completeness; independent tracing is not new [3]. Chan et al. supply the identity and infrastructure context [4,5]. Yun et al. already distinguish correct hidden-record guesses from supported findings and include abstention and a deterministic comparator [6]. Their binding-by-identifier-renaming factorial makes naming changes alone weak novelty. We adopt these distinctions rather than claim them as contributions.

Syscon captures supervisor-side execution receipts and reports that improved observability did not automatically improve detection [17]. ActionLens evaluates additional environment evidence and ablates its components [18]. These precedents require us to separate gains from recording more information from gains in a reader using the same information. Receipts and environmental evidence are not inventions of this project.

Who&When defines failure attribution in terms of a decisive error; BLADE audits information use at handoffs; BRIDGE evaluates verified runtime repair [19,20,10]. Our executor label does not answer those causal-responsibility or repair questions. HELIOS observes internal activations [7], while AI Control and CaMeLs motivate explicit trust and enforcement boundaries [8,9]. We do not claim to outperform these methods on their tasks.

Kitaoka et al. evaluate optimized attacks split across agents and compare separate with joint trajectory monitoring [21]. Their primary paper, recovered through the attached talk chat, motivates group evidence but already establishes distributed-attack monitoring failures. OpenAI describes investigations combining transcripts with external telemetry [22]. Incident traces have uncertain identities, timing and outcomes [11]; they cannot provide our complete scoring oracle.

The supplied talks guide scope; published methods and clearly identified source versions support scientific claims.

3. Methods

Scoping method. On 6 November 2026 we examined the current sprint materials and expanded the initial three-winner sample with relevant control and incident-response projects [12–15,17,18]. We reviewed the supplied chat/transcript files, followed their research links, and sought primary alternatives for blocked papers. The accompanying source ledger distinguishes full-text sections, abstracts, transcripts and remaining access gaps. This is a targeted scoping review, not an exhaustive or representative survey of Apart projects.

Selection and relevance. We considered oversight under evidence loss, external delegation enforcement, compromised-agent spread and cheap-model proxy validity. Checkable labels, closest prior art and completion risk favor a narrow oversight study. This is a qualitative decision, not a forecast of winning. We define the outcome before the records or metrics: the executor of one designated unauthorized fake mutation, or NONE if it did not occur.

Prospective environment. One coordinator and three workers operate through a bounded fake-data interpreter. The host records original actions, immutable instance IDs and lineage; it is trusted for recording, not assumed to enforce the task policy. Display-name churn changes labels without adding actors, actions or capability. Selective loss is a scripted transformation of investigator exports, not observed agent coordination and not deletion of the protected collector. No model output executes as a shell command or reaches real services.

Evidence and observation. Execution receipts R contain opaque call handles, request IDs, operations and post-execution status/state changes. Separate bindings B connect handles to immutable instances. R must not duplicate the identity information removed by the B ablation; B alone cannot prove execution. A pre-execution request is not a receipt of success. Primary comparisons retain B and R before export corruption; their removal is secondary. Targeted and random loss remove the same number of records within an instrumentation condition. This does not equate token counts or semantic information, so tokens, output caps and cost must be reported separately. Compute deletion masks before churn and reuse them across churn states and prompts.

Answerability. For packet x, instrumentation I and public information S, enumerate the finite compatible-history set H(x,I,S) under a published grammar and projection. A definite label is supported only if H is nonempty and every history agrees on that label; several labels require ABSTAIN, including when a guess happens to match hidden reality. Empty H means inconsistent evidence, reported and excluded from scoring. Primary condition/mask labels are latent; H takes the union over allowed settings while respecting every visible field. All readers receive the same public task rules, without leaked condition labels. This concerns a declared finite model, not all possible real-world histories.

Prospective comparisons. The main contrast is targeted minus random unsupported executor attribution at 25% loss with churn present, for one frozen base model and the plain primary

prompt. The difference-of-differences across churn conditions is secondary. Plain and evidence-constrained prompts receive byte-identical packets; a same-packet deterministic solver, always-abstain and fixed-actor baselines expose metric gaming. A full-host oracle is an upper bound. Name-permutation and denied/missing-result controls check the observation contract. The fixed deletion selector is not an optimized adaptive adversary.

Sampling and metrics. The costed proposal has 600 independent source episodes after 20 excluded pilots: 300 clean and 300 scripted violations across three families, with one family held out from prompt development. Six core packets, two prompts and one response for each of three candidate models imply 21,600 core calls. Two extra repetitions on 60 preselected episodes add 4,320 reliability calls; pilots add 720. The primary uses each first response on one frozen model/plain prompt; repeats and replication models are secondary. Pilot affordability and throughput must precede freezing. Use paired, stratified episode-bootstrap intervals, preserving complete pairs and reporting inconsistent-packet exclusions. Report structural answerability, unsupported accusations, invalidity, grounded correctness, supported definite coverage and cost. A retry is one planned call with its terminal status; INVALID remains in the scorable-call denominator. A handoff-free control tests whether effects exceed generic identifier sensitivity.

4. Results

Scoping finding. Selective evidence loss remains a focused candidate only if its interaction with identity churn yields information beyond the existing receipt, binding and failure-attribution literature. The synthesis identifies this candidate comparison, not a new detector or a measured defense gain. If the finite-history oracle cannot be validated promptly, the experiment should be narrowed before data collection. Table 1 identifies the closest precedents.

Table 1. Existing results and the narrower proposed test. These are research-design comparisons, not measured performance.

Prior work	**Already established**	**Proposed test**
Qin [3] / Yun [6]	Trace integrity; evidence support and abstention	Declared selective loss crossed with name churn
Syscon [17] / ActionLens [18]	External receipts; component ablations	Separate retained bindings from execution receipts
Who&When [19] / BLADE [20]	Decisive errors; handoff responsibility	Executor support within a bounded history model

Analytical observation. Fix all public rules and side information. Suppose two allowed histories H_A and H_B are equally likely and produce exactly the same complete visible packet O, but have different executors A and B. A reader restricted to O has the same response distribution in both. If

q_A and q_B are its probabilities of naming A and B, mean factual naming accuracy is $(q_A+q_B)/2 <= 1/2$, with equality whenever all response mass names A or B. Abstention avoids an unsupported accusation but supplies no definite executor. Unequal prior probabilities change the best guessing bound to the larger prior. This elementary construction is not a new general theorem and does not show that every deletion creates ambiguity. Both histories must remain compatible with the same public information; only abstention is justified for that ambiguous packet.

Hidden history	**Visible packet after erasure**	**Correct actor**
H_A: worker A updates record	Update complete; actor link absent	A
H_B: worker B updates record	Update complete; actor link absent	B

Figure 1. Two histories collapse to identical evidence. Any monitor restricted to this packet must guess or abstain. Retaining a trustworthy actor–operation receipt makes the observations differ. This construction is analytical, not an observed agent run.

An authentic receipt plus its instance binding can distinguish these particular histories when the differing relation is actually preserved. It need not resolve other ambiguities or establish the upstream cause of the mutation. Identical self-reported explanations cannot break the constructed ambiguity. Consequently, low accusation rates must be read together with useful definite coverage: an always-abstain reader must not be described as a successful detector.

There are no experimental agent runs, monitor accuracy estimates, fine-tuning results or measured defense gains in this report. The comparative result is the scoping decision and the analytical result is the bounded ambiguity construction. The empirical protocol is prospective.

5. Discussion and Limitations

The practical goal is defensible incident reconstruction: preserve the relations needed to establish an executed operation without falsely accusing an instance. Equal-packet comparisons isolate reader behavior; instrumentation changes test evidence availability. Neither establishes deployment security. Cooperation, collective agency, safety and executor attribution must remain distinct concepts [23–25].

The additional winning-project review sharpened the design rather than just its presentation. Syscon shows that better receipts can coexist with a null detection result; ActionLens motivates component ablations and usefulness costs. The original examples emphasize clean controls, causal checks, validation and restrained conclusions [13–15,17,18]. These are transferable design lessons, not evidence that a particular recipe guarantees a prize.

The central assumption is a correct finite history grammar and complete observation projection. Missing metadata, a hidden loss-policy hint or leaked identity can invalidate answerability labels. The same-packet deterministic solver is an exact reference only inside that bounded model. Real host compromise, off-gateway activity and unbounded histories are outside scope. The scripted export corruption does not establish spontaneous collusion, an attack on an immutable collector or real-world attack prevalence.

Novelty remains conditional on a direct comparison with the closest work. A generic logger, external receipt, abstention option or distributed-attack demonstration would duplicate existing contributions. The next step is independent oracle validation and a cost pilot, followed by a frozen protocol before confirmatory inference. Optional training must not contaminate the untouched test set. No empirical result can be supplied by editorial revision alone. Enumerating the minimum authentic evidence needed to restore supported attribution, against random add-back, could provide a useful preservation artifact. This is an untested diagnostic proposal, not a new runtime repair system.

6. Conclusion

The strongest defensible scope is an evaluation of supported executor attribution under a specified export-loss model and changing names. This review provides its primary-source rationale and a bounded ambiguity argument. The proposed multi-agent interaction, preservation tradeoff and practical monitor performance still require experiments.

Code and Data

Research artifacts and prospective protocol: https://github.com/razzyrazzy11/swarm-alignment . No experimental dataset has been collected. The October trusted-agent-sting project is preexisting work; its code and results are not new sprint contributions and have not been imported. Its prospective analysis and reproducibility practices informed this proposal [16].

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Scope and validity. This is a scoping study and prospective protocol, with one elementary analytical construction. No empirical safety guarantee follows. Literature coverage and ranking are selective; several organizer-linked papers could not be fully accessed. The assumed trusted host and closed tool interpreter require implementation and verification. A hash chain without an external anchor does not prevent full-history rewriting; even anchored integrity does not establish recording completeness.

Statistical limits. Define one false-accusation event over each independent clean episode's two primary calls. If all 300 clean episodes are scorable and have zero such events, the one-sided 95% upper bound is 1—0.05^(1/300), approximately 0.994%. It is exact for independent common-risk Bernoulli events and conservative for mean independent heterogeneous risk. The earlier 60-clean-episode profile gives 4.87%; at least 299 zero-event independent episodes are needed for below 1%. Recompute actual eligible n after exclusions. Repeated calls do not increase independent n, and the balanced cohort mixture is not deployment prevalence.

Potential misuse. The study could reveal which records an adversary should suppress or which identity changes confuse oversight. It could also help an operator falsely accuse a peer. Keep attack examples confined to fake-record simulations, publish defensive schemas and aggregate results, and review operational details under Apart's responsible disclosure policy before release. No live services, real personal records or incident exploit payloads are needed.

Evidence ethics. Distinguish attempted actions from successful execution and missing evidence from evidence of innocence. Do not assign intent from names, prose or agreement alone. Record uncertain labels, preserve benign provenance and allow insufficient-evidence outputs. Real incident samples require separate permission and redistribution checks. No human-subject study or operational testing is reported.

Appendix B. Prospective resource and release plan

Published Adaption rates imply a prospective 252.36-credit cost for 26,640 calls, assuming 6,000 input and 2,048 output tokens each [26]. This is a planning estimate before retries/secondary tests; account eligibility, balance and dollar conversion remain unverified. Reserve 125 of the proposed 500-credit envelope. No research credits have been spent. Pilot usage, throughput and output completeness; freeze models/prompts before confirmation. See the repository's ADAPTION_PLAN.md for dated rates, models, costing and optional-training constraints.

LLM Usage Statement

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